

Disease Model ER Diagram – Complete Documentation

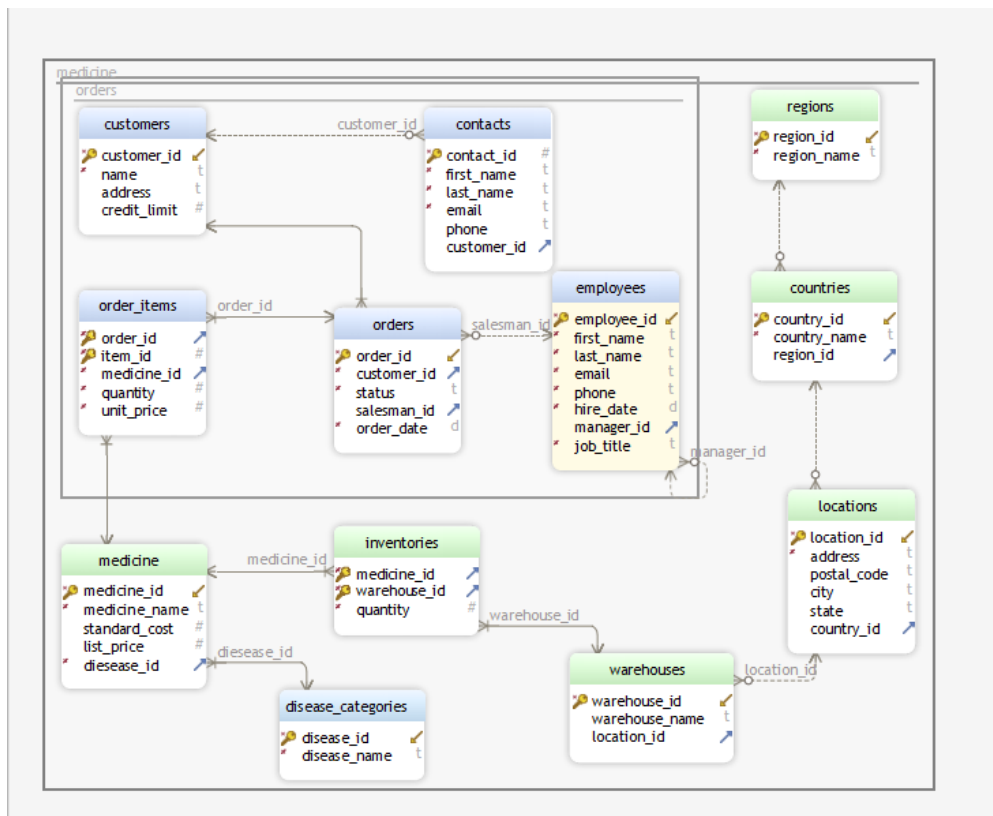
1. System Overview

This database represents a pharmaceutical sales and distribution system. It manages medicines, disease categories, customers, employees, warehouses, and order processing. The system ensures efficient tracking of inventory, customer purchases, and supply chain operations across multiple regions and locations.

Objectives

- Manage customer information and their orders
- Maintain medicine and disease mapping
- Track inventory across warehouses
- Enable order processing and sales tracking

2. ER Diagram



3. Entity Relationships

Key Relationships

- One **customer** can place many **orders** (1:M)
- One **order** contains many **order_items** (1:M)
- One **medicine** belongs to one **disease_category** (M:1)
- One **warehouse** stores many **medicines** via **inventories** (M:N resolved)
- One **employee** can manage many employees (self-referencing)
- One **region** contains many **countries** (1:M)
- One **country** contains many **locations** (1:M)
- One **location** contains many **warehouses** (1:M)

4. Cardinality & Optionality

- A customer may have zero or many orders
- An order must have at least one order item
- A medicine must belong to exactly one disease category
- A warehouse must have one location
- An employee may or may not have a manager
- Inventory must reference both a valid medicine and warehouse

5. Business Rules

- Each customer must have a unique customer_id
- Each order must be associated with a valid customer
- An order must contain at least one item
- Inventory quantity cannot be negative
- Each medicine must belong to one disease category
- Each warehouse must belong to one location
- A salesman (employee) must exist for each order
- Email addresses should be unique per contact

6. Data Dictionary Enhancements

Constraints Examples

- quantity INTEGER NOT NULL CHECK (quantity >= 0)
- email VARCHAR(255) UNIQUE NOT NULL
- customer_id INTEGER NOT NULL
- order_id INTEGER NOT NULL

7. Data Dictionary

Table employees

Idx	Field Name	Data Type
*🔑	employee_id	integer
*	first_name	varchar(255)
*	last_name	varchar(255)
*	email	varchar(255)
*	phone	varchar(50)
*	hire_date	date
🔗	manager_id	integer
*	job_title	varchar(255)
Indexes		
🔑	employees_pkey	ON employee_id
Foreign Keys		
🔗	fk_employees_manager	(manager_id) → employees (employee_id)
Referring Foreign Keys		
🔗	fk_employees_manager	(employee_id) ← employees (manager_id)
🔗	fk_orders_employees	(employee_id) ← orders (salesman_id)

8. Normalization

The database is normalized up to Third Normal Form (3NF):

- Eliminates redundancy by separating entities (customers, orders, medicines)
- Ensures each table represents a single concept
- Uses foreign keys to maintain relationships

9. Assumptions

- One medicine treats only one disease
- One order is handled by one salesman
- Customers have a single primary address
- Inventory is tracked per warehouse per medicine
- Each order item refers to one medicine only

10. Naming Standardization

Corrections required:

- Replace "disease" with "disease"
- Maintain consistent naming (snake_case recommended)

11. Derived Fields

- Total Order Cost = $\text{SUM}(\text{quantity} * \text{unit_price})$
- Inventory Value = $\text{quantity} * \text{standard_cost}$

12. Use Cases / Sample Queries

Example Queries

1. Find medicines for a disease

```
SELECT medicine_name  
FROM medicine  
WHERE disease_id = ?;
```

2. Customer order history

```
SELECT *  
FROM orders  
WHERE customer_id = ?;
```

3. Inventory by warehouse

```
SELECT warehouse_id, medicine_id, quantity  
FROM inventories;
```

4. Top-selling medicines

```
SELECT medicine_id, SUM(quantity) AS total_sold  
FROM order_items  
GROUP BY medicine_id  
ORDER BY total_sold DESC;
```

13. Indexing Strategy

- Primary keys are indexed for fast lookup
- Foreign keys are indexed for efficient joins
- Frequently queried columns (customer_id, medicine_id) should be indexed

14. Security & Access Control

- Admin: Full access to all tables
- Salesman: Insert orders, view customers
- Inventory Manager: Update inventories
- Read-only users: Reporting access

15. Metadata

- Author: [Your Name]
- Date: [Insert Date]
- Version: 1.0
- Tool Used: DbSchema

16. Conclusion

This ER model provides a structured and scalable design for managing pharmaceutical sales and distribution. It ensures data integrity, reduces redundancy, and supports efficient querying for business operations.